



ECHO Expedited Constrained Hierarchical Optimization

Routine Clinical Use of Clinical-Criteria–Driven Automated IMRT & VMAT Planning

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Conflict of Interest Disclosure



I am a co-inventor of ECHO (Expedited Constrained Hierarchical Optimization), the automated treatment-planning system presented in this talk.

ECHO's underlying technology has been licensed by MSK to two commercial vendors:



RaySearch Laboratories



Lumonous



Project Goal

Goal: Generate high-quality, consistent IMRT & VMAT plans **automatically** — directly inside the clinical workflow.



Fully Automated

Produces IMRT & VMAT plans driven directly by clinical criteria — with minimal human interaction.



No Prior Plans Needed

Optimization-based, not data-driven: no atlas, plan library, or training cases. Generalizes to any anatomy.



Native Eclipse Integration

Runs inside Eclipse— but powered by ECHO's own optimization engine, not Eclipse's.



The ECHO Workflow



Fully automated, end-to-end. Only Step 3 runs on the MSK server — every other step happens inside Eclipse. Planner effort is limited to setup and final review.



Two Key Ideas Behind ECHO

ECHO = Expedited **C**onstrained **H**ierarchical Optimization



1 · Constrained Optimization

Traditional — weighted sum:

$$\min w_{PTV} F_{PTV}(x) + w_{OAR} F_{OAR}(x)$$

Weights must be tuned by trial-and-error.

ECHO — hard constraints:

$$\min F_{PTV}(x)$$

$$\text{s. t. } F_{OAR}(x) \leq 20 \text{ Gy}$$

→ Clinical limits are **guaranteed**, not tuned.



2 · Hierarchical Optimization

Solve clinical priorities in sequence:

$$\textbf{Step 1: } F_{PTV}^* = \min F_{PTV}(x)$$

$$\text{s. t. } F_{OAR}(x) \leq 20 \text{ Gy}$$

Reach the best possible target coverage.

$$\textbf{Step 2: } \min F_{OAR}(x)$$

$$\text{s. t. } F_{PTV}(x) \leq 1.1 F_{PTV}^*$$

Then push OAR sparing further.

→ Target first, **then** maximal OAR sparing.

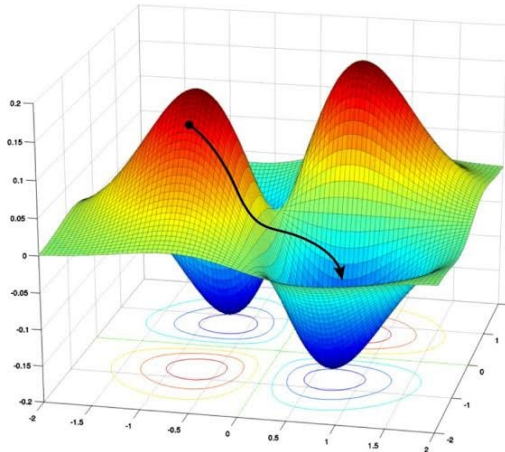


Extending ECHO to VMAT

Two challenges unique to VMAT — each addressed in a dedicated publication.



Non-Convex Optimization



Physics in Medicine & Biology



PAPER

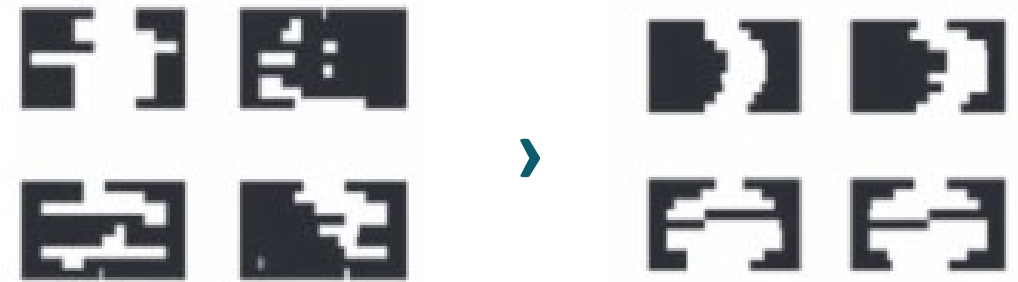
Solving the volumetric modulated arc therapy (VMAT) problem using a sequential convex programming method

Pinar Dursun, Masoud Zarepisheh, Gourav Jhanwar and Joseph O Deasy
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Keywords: volumetric modulated arc therapy, external radiation therapy, direct machine parameter optimization, sequential convex programming



Plan Complexity & Deliverability



Physics in Medicine & Biology



PAPER

Automated VMAT treatment planning using sequential convex programming: algorithm development and clinical implementation

Pinar Dursun*, Linda Hong, Gourav Jhanwar, Qijie Huang, Ying Zhou, Jie Yang, Hai Pham, Laura Cervino, Jean M Moran, Joseph O Deasy and Masoud Zarepisheh

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Simple to Use in the Clinic

1

Create a template plan

Set up structures & clinical criteria in Eclipse.

2

Run the ECHO plug-in

A single click — no manual weight tuning.

3

Plan is ready in Eclipse

Optimized plan returns for review & approval.

One click › ~30 min › plan ready. *No manual optimization.*

ECHO (Expedited Constrained Hierarchical Optimization) - Ver. 6.3.0.4

Email:

Disease Site:

View/Edit Targets

Previous Radiation
☐ No previous radiation
☒ With previous radiation

Run ECHO with user-defined Jaws: ☐

Advanced Parameters **Run Optimization** **Quit** **Save**

		Max Dose	Vol. (cc)	Res. (mm)
Z_PTV2mm	[Select]		434	
PTV	PTV		355	4
CTV	CTV		263	4
GTV	GTV		164	3.5
Bowel	[Select]		2956	
Bowel_Lg	Bowel-Lg	33 Gy	964	3.0
Bowel_privrt	Bowel-Sm-Treated	25 Gy	1236	3.0
Bowel_Sm	Bowel-Sm	33 Gy	2017	3.0
Cauda	Cauda	35 Gy	15	1.0
Cauda_privrt	Cauda-Treated	20 Gy	6	1.0
Cord	Cord	28 Gy	5	1.0
Kid_L	Kid_L	105.0 %	86	7.5
Kid_R	Kid_R	105.0 %	91	7.5
Kidneys	Kidneys	105.0 %	177	7.5
Patient Surface	Patient Surface		55045	
Skin	[Select]		1125	
ThecalSac	[Select]		38	
Z_CDR	[Select]		2958	
Z_overlap	[Select]		0	

The ECHO plug-in, running inside Eclipse



Clinical Impact at MSK

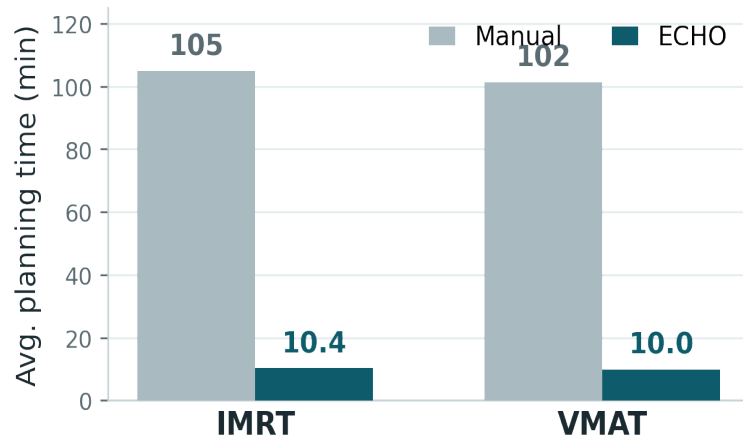
12,000+

patients treated with ECHO at MSK

Paraspinal · oligometas · prostate · lung · complex re-treat spine



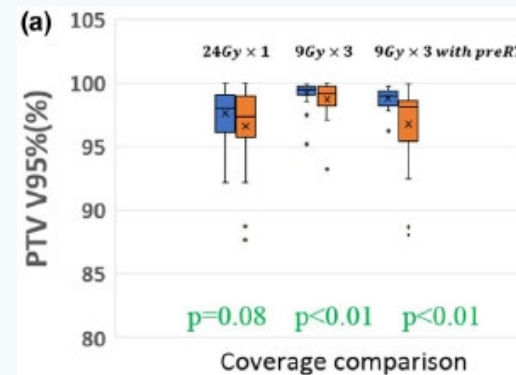
Saved time & resources



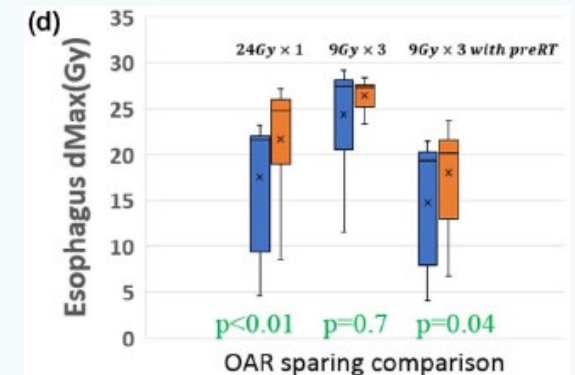
≈ 10× faster ~100 min → ~10 min per plan



Improved quality & consistency



Target coverage (PTV V95%)



OAR sparing (Esophagus Dmax)



ECHO matches target coverage while significantly improving OAR sparing (*p-values shown*).



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Summary



ECHO is an in-house automated planning system

Clinically deployed and integrated with Eclipse via API.



Powered by constrained + hierarchical optimization

Clinical-criteria-driven — no prior plans or training data needed.



12,000+ patients treated at MSK

Paraspinal, oligometas, prostate, lung — including complex re-treat spine.



Available now in RayStation; Eclipse coming soon

In clinical use for RayStation users today — Eclipse support arriving via the MSK–Lumonus collaboration.



Being open-sourced in Python via PortPy

Bringing ECHO's methods to the broader research community.



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Code available open-source via PortPy · github.com/PortPy-Project

Thank you!